

Hodge-relative completions of characteristic brane couplings

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Abstract

Chern–Simons theory is sometimes described, loosely, as Yang–Mills with the metric removed: a characteristic form is transgressed one degree down and the Hodge star is discarded, trading a locally dynamical quantity for a topological one. We ask what happens if this step is run in reverse: given a closed, gauge-invariant *characteristic* form on a D-brane worldvolume, does putting the metric back in – by forming its Hodge square – ever produce a genuinely new local coupling, one not already present in the Dirac–Born–Infeld (DBI) or Wess–Zumino (WZ) sectors? We show that, to first order in a change of representative, the obstruction is precisely the codifferential $\delta_g P$; representative-independence at this order holds exactly on the co-closed locus. We prove this as a necessary and sufficient condition, with an explicit formula for the obstruction in terms of the codifferential. We then apply the resulting bookkeeping to the D4-brane at leading order and prove a small, honest no-go theorem: every Hodge-completed coupling built from the available field content at this order already belongs to a known sector – DBI, ordinary flux kinetic terms, or characteristic descent – and the one pairing that would mix the two generators fails for pure form-degree reasons. We close by identifying, through the same bookkeeping, the first point at which brane extrinsic geometry could produce a genuinely mixed class, and pose this as a precise open problem.

1 Introduction

There is a familiar sense in which Chern–Simons theory is ‘Yang–Mills without a metric.’ Given a gauge-invariant characteristic polynomial $P(F)$, closed by the Bianchi identity, with local transgression $P(F) = dQ_{\text{CS}}(A)$, the functional $\int Q_{\text{CS}}(A)$ never calls on a Riemannian structure: it is built from A , F , and the wedge product alone. The Yang–Mills functional $\int \text{Tr}(F \wedge *_g F)$, by contrast, exists only once a metric has been chosen, and it is exactly that dependence which turns a quantity sensitive only to a bundle’s characteristic classes into a genuinely local field theory, with a stress tensor, propagation, and scattering. We take this as a slogan rather than a theorem: the two functionals are related by transgression *and* by deleting a metric, two logically separate operations, and we are careful below not to conflate them.

The question we actually ask is the reverse one, stated without reference to any particular pair of actions: *given a closed, gauge-invariant form built from the characteristic data of a D-brane – curvatures, fluxes, Ramond–Ramond field strengths – when does reinstating a worldvolume metric, by forming its Hodge square, produce a local coupling that is not already accounted for by the standard DBI and WZ sectors?* A first guess is that this should be governed by de Rham cohomology: forms in the same class ought to complete to the same local theory. This guess is false, and its failure is not a technical nuisance but the entire content

of the problem: a fixed cohomology class remembers only global, bundle-level information, while its Hodge square is manifestly a function of the chosen representative, and the gap between the two is precisely the space in which new local physics could live.

Making this precise requires some care, and three clarifications are worth stating before any formula appears. First, the relevant ambiguity in a representative is *not* a gauge transformation of a connection: for $F = dA$, shifting A by an exact one-form leaves F untouched, so there is no room here for new physics. The ambiguity we study instead arises for genuinely closed characteristic forms such as the Neveu–Schwarz field strength $H = dB$, whose cohomology class is fixed but whose specific representative – the choice of *curving* on the underlying gerbe – is extra, physical data not captured by $[H]$ alone; this is precisely the content of differential cohomology [4, 5], and it is the correct and rigorous setting for the question we ask. Second, ‘topological’ has to be defined narrowly: we reserve it for forms that are exact *within the characteristic algebra itself*, in the spirit of Chern–Simons descent, rather than for the much larger and less informative class of forms with vanishing Euler–Lagrange variation. Third, we make no claim that any of the individual algebraic ingredients below are new; the variational bicomplex and local BRST cohomology already supply the correct home for local Lagrangians modulo total derivatives and gauge redundancy [1, 2], and the classification of higher-derivative D-brane operators, including

terms quadratic in H , is already extensive and tightly constrained by T-duality [8, 7, 6, 9]. What has not, to our knowledge, been isolated is the precise criterion governing when a metric completion of characteristic data is representative-independent, and the use of that criterion as a classification tool.

Our results are two small, self-contained statements. Theorem 3.2 identifies that criterion exactly: to first order in a change of representative, the obstruction is precisely the codifferential $\delta_g P$, and representative-independence at this order holds if and only if the representative is co-closed. Theorem 4.2 then applies this bookkeeping to the D4-brane and shows that its entire leading-order Hodge-completed sector, with no exceptions, already belongs to DBI, ordinary flux dynamics, or characteristic descent; the one candidate that would genuinely mix the two available generators does not exist, for reasons of form degree alone. This is a calibration result: it tells us exactly how far the obvious search space is from producing anything new, and Section 5 uses the same bookkeeping to say precisely where a nontrivial class could first appear.

2 The relative completion problem

Fix an oriented worldvolume W of dimension d with metric g , and a field bundle $E \rightarrow W$ carrying the variational bicomplex splitting $d = d_H + d_V$ on its infinite jet bundle [1]. A local Lagrangian is a horizontal top form $L \in \Omega_{\text{loc}}^{d,0}(J^\infty E)$, with Euler–Lagrange form defined through $d_V L = \text{EL}(L) + d_H \Theta_L$.

Definition 2.1. Let $\mathcal{C}_{\text{ext}} = \langle \phi^* \Omega^\bullet(X), \mathcal{F}, H, R, D \rangle_{\wedge, d, D}$ be the algebra of gauge-invariant forms built covariantly from the pulled-back bulk fields and the worldvolume field strength, using no metric. Its descent sector is

$$\mathcal{W}^d = \{ \tau \in \mathcal{C}_{\text{ext}}^d : \tau = d\sigma, \sigma \in \mathcal{C}_{\text{ext}}^{d-1} \},$$

the elements exact within $\mathcal{C}_{\text{ext}}$ itself – the natural home of Chern–Simons/Wess–Zumino descent. The metric completion is $\mathcal{C}_H(g) = \langle \mathcal{C}_{\text{ext}}, *_g \rangle_{\wedge, d, D, *_g}$, obtained by adjoining the Hodge star.

Definition 2.2. The Hodge-relative variational coupling space in degree d is

$$\mathfrak{V}_{H/\text{Top}}^d(g) = \frac{\mathcal{C}_H^d(g)}{\ker(\text{EL}) + \mathcal{W}^d + \mathcal{R}^d}, \quad (1)$$

where $\mathcal{R}^d$ collects the standard redundancies of the effective description: Bianchi and algebraic identities, integration by parts, chosen field redefinitions. By the triviality theorem for the variational bicomplex,

$\ker(\text{EL})$ coincides locally with $d_H \Omega_{\text{loc}}^{d-1,0}$ [1], so this recovers the expected statement that boundary terms are discarded.

Figure 1 depicts the quotient of Definition 2.2. A nonzero class in $\mathfrak{V}_{H/\text{Top}}^d(g)$ is, by construction, a local coupling with nonvanishing Euler–Lagrange variation that is neither a boundary term nor a characteristic/WZ coupling nor removable by the identities in $\mathcal{R}^d$: exactly the sense in which a Hodge completion can fail to be redundant.

3 The completion theorem

Fix a class $[P] \in H_{\text{dR}}^p(W; \mathbb{R})$ represented by a closed form $P \in \mathcal{C}_{\text{ext}}^p$, $dP = 0$. As discussed in the Introduction, we do *not* interpret a shift of representative as a gauge transformation of an underlying potential; we interpret it as the choice of differential-cohomological refinement – the curving – of a fixed class, which for the Neveu–Schwarz flux is exactly the freedom $H \mapsto H + d\mu$ by a globally defined two-form μ , corresponding to a different choice of B -field profile with the same periods [4, 5].

Definition 3.1. For $P \in \mathcal{C}_{\text{ext}}^p$, the Hodge completion of P is $K_g(P) = P \wedge *_g P$. Since $*_g P \in \Omega^{d-p}$, this is always a top-degree form, $K_g(P) \in \Omega^d(W)$, regardless of p .

Theorem 3.2 (Completion criterion). *Let W be closed and oriented, and let δ_g denote the codifferential of g . For any $(p-1)$ -form μ ,*

$$K_g(P + d\mu) - K_g(P) \equiv 2\varepsilon \mu \wedge *_g \delta_g P + d\mu \wedge *_g d\mu \quad (2)$$

modulo d_H -exact terms, for a sign $\varepsilon = \varepsilon(p, d) = \pm 1$ fixed once and for all by the Hodge-star and codifferential conventions in use (e.g. [3], §7). Consequently, as a class in $\mathfrak{V}_{H/\text{Top}}^d(g)$, the completion $[K_g(P)]_{\text{rel}}$ is independent of the choice of representative within $[P]$, to linear order in the shift, for every shift μ **if and only if** $\delta_g P = 0$: that is, if and only if P is co-closed, which together with $dP = 0$ means P is g -harmonic.

Proof. Expand $K_g(P + d\mu) - K_g(P) = 2 d\mu \wedge *_g P + d\mu \wedge *_g d\mu$, using bilinearity of $\wedge$ and $*_g$ and the pointwise symmetry $\alpha \wedge *_g \beta = \beta \wedge *_g \alpha$ for equal-degree forms. For the linear term, Stokes’ theorem on the closed manifold W gives, as an identity of local densities modulo an exact top form,

$$d\mu \wedge *_g P \equiv (-1)^p \mu \wedge d *_g P \pmod{d(\mu \wedge *_g P)},$$

and the correction term $d(\mu \wedge *_g P)$ is exact, hence lies in $d_H \Omega_{\text{loc}}^{d-1,0} \subset \ker(\text{EL})$ and is discarded by Definition 2.2. The standard Hodge-theoretic identity relating the exterior derivative of a Hodge dual to the

$$\ker(\text{EL}) + \mathcal{W}^d + \mathcal{R}^d \hookrightarrow \mathcal{C}_H^d(g) \xrightarrow{\pi} \mathfrak{V}_{H/\text{Top}}^d(g)$$

Figure 1: The quotient defining the Hodge-relative variational coupling space: the discarded ideal (left) embeds in the completed algebra, which surjects onto the coupling space via π .

$$\begin{array}{ccc} P & \xrightarrow{K_g} & K_g(P) \\ \downarrow [\cdot]_{\text{dR}} & & \downarrow [\cdot]_{\text{rel}} \\ [P] & \dashrightarrow & [K_g(P)]_{\text{rel}} \\ & \mathrel{\mathop{\neq}} & \end{array}$$

Figure 2: The completion square of Theorem 3.2. The top map and both vertical quotients always exist; the dashed bottom map exists only along the harmonic locus $\delta_g P = 0$.

codifferential, $d *_g P = \varepsilon' *_g \delta_g P$ for a sign $\varepsilon' = \varepsilon'(p, d)$ fixed by convention [3], then gives Equation (2) with $\varepsilon = (-1)^p \varepsilon'$. This vanishes identically, for every choice of μ , precisely when $\delta_g P = 0$ as a form (a nonzero $\delta_g P$ can always be paired against some compactly supported μ to give a nonvanishing density, since $\mu \wedge *_g \delta_g P = \langle \mu, \delta_g P \rangle_g \text{vol}_g$ is the pointwise Hodge inner product). Together with the given closure $dP = 0$, $\delta_g P = 0$ is exactly the Hodge–de Rham harmonicity condition. $\square$

Corollary 3.3. *The completion map K_g descends, to linear order in the representative, to a well-defined function of $[P] \in H_{\text{dR}}^p(W)$ if and only if P is evaluated in the (unique, metric-dependent) harmonic representative of its class. For any other representative, the representative-dependence of K_g is entirely governed by $\delta_g P$: the failure of P to be co-closed is the obstruction.*

Figure 2 summarises the picture: the top map is always available and the two vertical maps are the evident quotients, but the dashed bottom map exists – by Theorem 3.2 – only along the harmonic locus. This is the precise sense in which the Hodge star cannot turn topology into dynamics for free: it takes a representative, not a class, and the price of forgetting the representative is exactly the codifferential obstruction above. It is also, we think, a more informative statement than the vaguer assertion that ‘Hodge completion is representative-dependent’: it says exactly which representatives are safe, and exactly what controls the failure for the rest.

4 A leading-order no-go theorem on the D4-brane

We now put Definition 2.2 to work on the smallest brane for which a genuinely mixed completion is dimensionally conceivable. Let W_5 be the worldvolume of a D4-brane, $\dim W_5 = 5$, and let $\mathcal{F} = 2\pi\alpha' F + \phi^* B$ be the gauge-invariant worldvolume two-form, with $d\mathcal{F} = \phi^* H$, $H = dB$ the closed, gauge-invariant NS flux of Theorem 3.2. At the lowest order – one factor of each available closed generator, at most one Hodge insertion, no further derivatives – there are exactly four candidate top-degree densities, and their fate is settled by nothing more than form-degree arithmetic together with one elementary computation.

Lemma 4.1. $\mathcal{F} \wedge H = \frac{1}{2} d(\mathcal{F} \wedge \mathcal{F})$. In particular $\mathcal{F} \wedge H \in \mathcal{W}^5$.

Proof. Since $\mathcal{F}$ has even degree, $d\mathcal{F} \wedge \mathcal{F} = \mathcal{F} \wedge d\mathcal{F}$, so $d(\mathcal{F} \wedge \mathcal{F}) = 2d\mathcal{F} \wedge \mathcal{F} = 2\phi^* H \wedge \mathcal{F}$. The claim follows. $\square$

Lemma 4.1 is worth pausing on: it says that the one pairing of $\mathcal{F}$ and H that exists *without any metric at all* is not merely quasi-topological in spirit, it is a literal total derivative, transgressed from $\frac{1}{2}\mathcal{F} \wedge \mathcal{F}$ exactly as a Chern–Simons density is transgressed from a characteristic polynomial. It is, in the cleanest possible sense, already spoken for.

Theorem 4.2 (Leading-order vanishing). *Within the leading-order ansatz consisting of one factor of each generator, at most one Hodge insertion, and no additional derivatives, the only degree-five monomials are $\mathcal{F} \wedge H$, $\mathcal{F} \wedge *_5 \mathcal{F}$, and $H \wedge *_5 H$. These lie respectively in $\mathcal{W}^5$, the DBI kinetic sector, and the ordinary flux-kinetic sector. There is no fourth, genuinely mixed class: the only pairing that would combine one factor of each generator with a single Hodge insertion, $\mathcal{F} \wedge *_5 H$, has degree $2 + 2 = 4 \neq 5$ and simply does not define a Lagrangian density on W_5 . Consequently*

$$\mathfrak{V}_{H/\text{Top}}^5(g) = 0$$

at this order: the naive characteristic-to-Hodge completion problem produces no new interaction on the D4-brane.

Proof. Immediate from Table 1: $\mathcal{F} \wedge H \in \mathcal{W}^5$ by Lemma 4.1, and $\mathcal{F} \wedge *_5 \mathcal{F}$, $H \wedge *_5 H$ are, by inspection in local coordinates, proportional to $F_{\mu\nu} F^{\mu\nu} \text{vol}_g$ and

Candidate	Degree	Star insertions	Fate
$\mathcal{F} \wedge H$	$2 + 3 = 5$	0	$\mathcal{W}^5$ (Lemma 4.1)
$\mathcal{F} \wedge *_5 \mathcal{F}$	$2 + 3 = 5$	1	DBI kinetic term
$H \wedge *_5 H$	$3 + 2 = 5$	1	ordinary flux-kinetic term [8]
$\mathcal{F} \wedge *_5 H$	$2 + 2 = 4$	1	<i>not a top form</i>

Table 1: The complete leading-order candidate list on the D4-brane.

$H_{\mu\nu\rho}H^{\mu\nu\rho}\text{vol}_g$ respectively – the standard quadratic DBI term and an ordinary flux-kinetic term already present in the higher-derivative D-brane literature [8]. Since $*_5$ sends p -forms to $(5 - p)$ -forms, no other top-degree pairing of $\mathcal{F}$ and H with a single star exists. $\square$

The value of this result is therefore diagnostic rather than constructive: it shows that the simplest intrinsic D4 search space is exhausted before genuinely mixed dynamics can appear, with the emptiness traceable to nothing more exotic than form-degree bookkeeping. A search that had instead reported $H \wedge *_5 H$ as a new coupling would have been reporting a term already known from T-duality analyses of higher-derivative brane actions [8]; the actual content of Theorem 4.2 is the classification, not any single entry in it.

5 Where new physics could first appear

Theorem 4.2 used only the intrinsic worldvolume data $\mathcal{F}, H$. A D-brane, however, is also an embedded object, and its Wess–Zumino action is sensitive to the curvature R_T of the tangent bundle and R_N of the normal bundle through the characteristic ratio $\sqrt{\widehat{A}(TW)/\widehat{A}(NW)}$, together with the second fundamental form Ω , a one-form valued in the normal bundle. Enlarging $\mathcal{C}_{\text{ext}}$ to include these generators is the natural next step, and the same bookkeeping used above already shows where the leading-order vanishing of Section 4 must eventually break down.

The characteristic curvature invariants $\text{Tr}(R_T \wedge R_T)$ and $\text{Tr}(R_N \wedge R_N)$ are closed four-forms on W_5 by the Bianchi identity, exactly as H was closed by construction. Their Hodge squares have total degree $4 + 1 = 5$ and are the direct curvature analogues of $H \wedge *_5 H$: ordinary higher-derivative corrections, not new in kind. What Table 1 shows is that a genuinely mixed class needs a pairing of *different* generators at matching degree, and the smallest such pairing available at this enlarged order is between a four-form curvature invariant and a one-form: exactly the degree of Ω . A term of the schematic shape

$$\text{Tr}(R_T \wedge R_T) \wedge \iota_\Omega(\mathcal{F} \text{ or } H)$$

– a curvature invariant paired, through the extrinsic geometry, with the intrinsic flux data – is the first candidate that pairs characteristic (global, bundle-level) information with genuinely local (embedding-dependent) information in a single irreducible density, in the sense described at the end of Section 2. Whether such a pairing exists as a nonzero, gauge-covariant, $\mathcal{W}$ -independent contraction, and whether it survives κ -symmetry and T-duality, is a well-posed question that we do not resolve here.

Open Problem 5.1. *Extend $\mathcal{C}_{\text{ext}}$ by R_T, R_N, Ω . Determine the minimal total derivative order $n_\star$ at which $\mathfrak{V}_{H/\text{Top}}^5(g)$, computed with this enlarged generator set, first contains a class not already reducible to DBI, ordinary curvature- or flux-kinetic terms, or $\mathcal{W}^5$. Theorem 4.2 shows $n_\star > 0$ (using $\mathcal{F}, H$ alone); the discussion above suggests the first candidates appear once Ω is included, at the order of one curvature invariant plus one extrinsic-geometry contraction.*

6 Discussion

The reversal of the Yang–Mills→Chern–Simons reduction turns out to hinge on a single, checkable fact: a closed form and its Hodge square do not know about each other except through the codifferential, and the map from characteristic data to local dynamics is well-defined exactly on the harmonic locus and nowhere else (Theorem 3.2). On the D4-brane this abstract statement resolves into an unglamorous but honest fact: the naive search space is degree-obstructed before any dynamics enters, and every leading-order Hodge completion is already DBI, an ordinary flux-kinetic term, or characteristic descent (Theorem 4.2).

We see three natural directions from here, each precise enough to be either proved or disproved rather than merely gestured at: whether the extrinsic-geometry pairing of Open Problem 5.1 is nonzero and survives κ -symmetry [7]; whether the assignment $[P] \mapsto [K_g(P)]_{\text{rel}}$ is compatible with T-duality, which is already known to fix the coefficients of comparable flux- and curvature-squared operators [8]; and whether the same bookkeeping, applied at a brane boundary or defect where the relevant Bianchi identity acquires a

source, $dH = J$, lets a characteristic source term feed directly into local dynamics through $d(H \wedge *_g H) = J \wedge *_g H + \dots$, in the spirit of anomaly inflow [9]. None of these is claimed here; what is claimed is a clean criterion for when they could possibly succeed, and a demonstration that the first, most obvious place to look does not.

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